Random variables

Topic 1: What is a Random Variable?

1. First understand a random experiment

A random experiment is an experiment whose outcome cannot be predicted with certainty before performing it.

Examples:

- Tossing a coin.
- Rolling a dice.
- Drawing a card.
- Selecting a student randomly.

Example: Tossing a coin twice

Sample space:

$$S = \{HH, HT, TH, TT\}$$

There are four possible outcomes.

2. Definition of Random Variable

A random variable is a function that assigns a numerical value to every outcome of a random experiment.

Mathematically:

$$X : S \to \mathbb{R}$$

Topic 2: Types of Random Variables

There are two important types.

A. Discrete Random Variable

A random variable is discrete if it takes a finite or countably infinite set of values.

Examples:

- Number of heads in 3 coin tosses: 0, 1, 2, 3.
- Number of students in a classroom.
- Number of defective products in a box.

B. Continuous Random Variable

A random variable is continuous if it can take any value within an interval.

Examples:

- Height of a student: 165.2 cm.
- Weight: 58.75 kg.
- Temperature: 32.6°C.
- Time taken to complete an exam: 120.5 minutes.

Difference between them

Discrete	Continuous
Countable values	Any value in an interval
Usually counting	Usually measurement
0, 1, 2, 3	1.2, 1.25, 1.256
PMF	PDF

Q1. A coin is tossed 4 times. X = Number of heads. What type of random variable is X?

- A. Discrete
- B. Continuous
- C. Both
- D. Neither

Q2. The weight of a randomly selected student is a:

- A. Discrete random variable
- B. Continuous random variable
- C. Constant
- D. None

Q3. A dice is rolled. X = Number appearing on the upper face. Which values can X take?

- A. 0, 1, 2, 3, 4, 5, 6
- B. 1, 2, 3, 4, 5, 6
- C. Any real number
- D. Only 1 and 6

Topic 4: Probability Mass Function (PMF)

1. What is PMF?

PMF stands for **Probability Mass Function**.

It gives the probability that a discrete random variable takes a particular value.

Mathematically:

$$P(X = x) = p(x)$$

Tinglish: Discrete random variable oka particular value teesukune probability ni PMF antaru.

3. Two important properties of PMF

Property 1: Non-negativity

$$P(X = x) \geq 0$$

Every probability must be zero or positive.

Property 2: Sum of all probabilities equals 1

$$\sum_{x} P(X = x) = 1$$

Q1. A random variable X has PMF: P(0)=0.2, P(1)=0.5, P(2)=0.3. Find P(X≥1).

- A. 0.2
- B. 0.3
- C. 0.8
- D. 0.5

Q2. Which of the following can be a valid PMF?

- A. 0.2, 0.3, 0.5
- B. 0.4, 0.4, 0.4
- C. -0.2, 0.5, 0.7
- D. 0.1, 0.2, 0.4

Q3. If P(X=0)=0.25, P(X=1)=0.35, P(X=2)=k, find k.

- A. 0.25
- B. 0.35
- C. 0.40
- D. 0.50

1. What is PDF?

PDF stands for **Probability Density Function**.

It describes how probability is distributed over the possible values of a continuous random variable.

Mathematically, if X is a continuous random variable with PDF $f(x)$:

$$f(x) \geq 0$$

Probability between two values is calculated using integration:

$$\boxed{P(a \leq X \leq b) = \int_a^b f(x)\,dx}$$

Property 1: Non-negativity

$$f(x) \geq 0$$

PDF cannot be negative.

Property 2: Total area equals 1

$$\boxed{\int_{-\infty}^{\infty} f(x)\,dx = 1}$$

Tinglish: PDF graph kinda total area compulsory 1 undali.

Property 3: Probability of a single point is zero

$$\boxed{P(X = a) = 0}$$

Q1. For a valid PDF, which condition must always hold?

- A. f(x) must be negative
- B. Total integral must equal 1
- C. f(x) must always equal 1
- D. Total integral must equal 0

Q2. If X is continuous, what is P(X = 5)?

- A. 1
- B. 0.5
- C. 0
- D. Cannot be determined

Q3. If f(x)=2x for 0≤x≤1, find P(0≤X≤0.5).

- A. 0.25
- B. 0.5
- C. 0.75
- D. 1

Q4. A PDF is f(x)=k for 0≤x≤4 and zero otherwise. Find k.

- A. 1
- B. 1/2
- C. 1/4
- D. 4

1. What is CDF?

CDF stands for **Cumulative Distribution Function**.

It gives the probability that a random variable X takes a value less than or equal to x.

Mathematically:

$$\boxed{F(x) = P(X \leq x)}$$

3. CDF for continuous random variables

For continuous random variables:

$$\boxed{F(x) = \int_{-\infty}^{x} f(t)\,dt}$$

Here, $f(t)$ is the PDF.

And:

$$P(a < X \leq b) = F(b) - F(a)$$

4. Four important properties of CDF

Property 1: Non-decreasing

$$x_1 < x_2 \Rightarrow F(x_1) \leq F(x_2)$$

CDF never decreases.

Property 2: Range

$$0 \leq F(x) \leq 1$$

Property 3: Limits

$$F(-\infty) = 0, \quad F(\infty) = 1$$

Property 4: Right-continuous

CDF is always right-continuous.

Q1. If F(2)=0.6 and F(5)=0.9, find P(2<X≤5).

- A. 0.3
- B. 0.6
- C. 0.9
- D. 0.5

Q2. Which statement about CDF is always true?

- A. Always decreasing
- B. Always non-decreasing
- C. Always constant
- D. Can be negative

Q3. If X is continuous, what is P(X=a)?

- A. F(a)
- B. 1-F(a)
- C. 0
- D. Always 1

Q4. For a CDF, what is F(∞)?

- A. 0
- B. 0.5
- C. Depends on PDF
- D. 1

Q5. A continuous random variable has $F(x)=x^2$ for $0 \le x \le 1$. Find $P(0.2 < X \le 0.6)$.

- A. 0.32
- B. 0.36
- C. 0.40
- D. 0.20

Section A – MCQs

Q1. A discrete random variable X has the following PMF:

X	0	1	2
P(X)	0.2	0.3	0.5

Find $P(X \geq 1)$.

A. 0.2

B. 0.3

C. 0.8

D. 1

Q2. A continuous random variable has PDF:

$$f(x) = kx, \quad 0 \leq x \leq 2$$

Find k.

A. 1/2

B. 1/4

C. 1

D. 2

Q3. If X is a continuous random variable, then $P(X = 3)$ is:

A. 1

B. 0.5

C. 0

D. Cannot be determined

Q4. Given the CDF:

$$F(x) = \begin{cases} 0 & x < 0 \\ x/4 & 0 \le x \le 4 \\ 1 & x > 4 \end{cases}$$

Find $P(1 < X \le 3)$.

A. 0.25

B. 0.5

C. 0.75

D. 1

Q5. Which of the following is always true for a CDF?

A. It is decreasing.

B. It is non-decreasing.

C. It is always differentiable.

D. It can exceed 1.

Q6. If a PDF is given by $f(x) = 2x$ for $0 \le x \le 1$, find $P(X > 0.5)$.

A. 0.25

B. 0.50

C. 0.75

D. 1

Section B – MSQs

Select ALL correct options

Q7. Which statements about PMF are correct?

A. Every probability must be non-negative.

B. Sum of all probabilities must equal 1.

C. PMF is used for continuous random variables.

D. Probability at a particular value can be positive.

Q8. Which statements about PDF are correct?

A. PDF must be non-negative.

B. Total integral over the real line must equal 1.

C. PDF value at a point is always the probability at that point.

D. Probability over an interval is calculated using integration.

Q9. Which statements about CDF are correct?

A. $F(x) = P(X \leq x)$

B. CDF is non-decreasing.

C. $F(\infty) = 1$

D. CDF can be negative.

Q10. A continuous random variable X has:

$$f(x) = \begin{cases} 2x & 0 \leq x \leq 1 \\ 0 & \text{otherwise} \end{cases}$$

Which statements are correct?

A. $P(X = 0.5) = 0$

B. $P(X \leq 0.5) = 0.25$

C. $P(X > 0.5) = 0.75$

D. $F(1) = 1$

1. What is expectation?

Definition: Expected value is the weighted average of all possible values of a random variable, where probabilities act as weights.

In simple words: A random variable can take different values. Expectation tells us its average value over many repeated experiments.

2. Formula for a discrete random variable

$$E[X] = \sum_{x} xP(X = x)$$

Meaning:

- x = possible value of random variable.
- $P(X = x)$ = probability of that value.
- Multiply each value by its probability and add everything.

3. Solved example – Fair die

A fair die is rolled once. Let X be the number appearing.

Possible values:

$$X = \{1, 2, 3, 4, 5, 6\}$$

Each value has probability $1/6$.

$$E[X] = 1\left(\frac{1}{6}\right) + 2\left(\frac{1}{6}\right) + \cdots + 6\left(\frac{1}{6}\right)$$

$$= \frac{1+2+3+4+5+6}{6} = \frac{21}{6} = \boxed{3.5}$$

Important: 3.5 is not a possible outcome of one die roll. It is the long-run average.

Topic 2: Properties of Expectation

These properties are very important for GATE DA.

Property 1: Expectation of a constant

$$E[c] = c$$

Example:

$$E[5] = 5$$

Property 2: Constant multiplication

$$E[aX] = aE[X]$$

Example: If $E[X] = 4$, then

$$E[3X] = 3(4) = 12$$

Property 3: Addition of a constant

$$E[X + c] = E[X] + c$$

Example: If $E[X] = 4$,

$$E[X + 5] = 4 + 5 = 9$$

Property 4: Linearity of expectation

$$E[X + Y] = E[X] + E[Y]$$

Important GATE point: **This property does not require X and Y to be independent.**

More generally:

$$E[aX + bY + c] = aE[X] + bE[Y] + c$$

Q1. A fair coin is tossed once. X = 1 for Head and 0 for Tail. Find E[X].

- A. 0
- B. 1/4
- C. 1/2
- D. 1

Q2. If E[X] = 5, find E[3X + 2].

- A. 15
- B. 17
- C. 10
- D. 7

Q3. For any two random variables X and Y, which statement is always true?

- A. E[X+Y] = E[X]E[Y]
- B. E[X+Y] = E[X]+E[Y]
- C. X and Y must be independent
- D. E[XY] = E[X]E[Y]

1. What is variance?

Variance measures how far the possible values of a random variable are spread around their expected value.

2. Formula

The most important GATE DA formula:

$$\boxed{Var(X) = E[(X - \mu)^2]}$$

Where:

- X = Random variable
- $\mu = E[X]$ = Mean or expected value
- $X - \mu$ = Deviation from mean

An equivalent and very useful formula:

$$\boxed{Var(X) = E[X^2] - (E[X])^2}$$

GATE shortcut Remember this second formula. It is often faster for numerical problems.

Property 1: Variance of a constant

$$Var(c) = 0$$

Property 2: Adding a constant

$$Var(X + c) = Var(X)$$

Adding a constant shifts all values but doesn't change their spread.

Property 3: Multiplying by a constant

$$Var(aX) = a^2 Var(X)$$

Property 4: General formula

$$\boxed{Var(aX + b) = a^2 Var(X)}$$

Remember: The constant b disappears, but a gets squared.

5. Standard deviation

Standard deviation is the square root of variance.

$$\boxed{\sigma = \sqrt{Var(X)}}$$

In our example:

$$\sigma = \sqrt{1} = 1$$

Q1. If $E[X] = 4$ and $E[X^2] = 20$, find Var(X).

- A. 4
- B. 16
- C. 20
- D. 36

Q2. If Var(X) = 5, find Var(3X + 7).

- A. 15
- B. 22
- C. 45
- D. 52

Q3. If Var(X) = 8, what is Var(X + 100)?

- A. 108
- B. 100
- C. 8
- D. 0

Q4. If X is a constant random variable equal to 7, its variance is:

- A. 7
- B. 49
- C. 1
- D. 0

Rule	Formula
Basic variance	$Var(X) = E[X^2] - (E[X])^2$
Constant	$Var(c) = 0$
Adding constant	$Var(X + b) = Var(X)$
Multiplying constant	$Var(aX) = a^2 Var(X)$
Combined	$Var(aX + b) = a^2 Var(X)$

Topic 4: Expectation of a function of X

Sometimes the question gives X, but asks us to calculate X^2, X^3, or another function of X.

The formula is:

$$\boxed{E[g(X)] = \sum_x g(x)P(X = x)}$$

Topic 5: Continuous random variable – Expectation and variance

For a continuous random variable with PDF $f(x)$:

Expected value:

$$\boxed{E[X] = \int_{-\infty}^{\infty} x f(x)\, dx}$$

Expected square:

$$E[X^2] = \int_{-\infty}^{\infty} x^2 f(x)\, dx$$

Variance:

$$\boxed{Var(X) = E[X^2] - (E[X])^2}$$

Topic 6: Important formula revision

Concept	Discrete	Continuous
Expectation	$\sum xf(x)$	$\int xf(x)dx$
Expected square	$\sum x^2p(x)$	$\int x^2f(x)dx$
Variance	$E[X^2] - E[X]^2$	$E[X^2] - E[X]^2$

Q1. If E[X] = 3 and $E[X^2]$ = 11, find Var(X).

- A. 2
- B. 9
- C. 11
- D. 20

Q3. For a continuous random variable, $E[X^2]$ is calculated using:

- A. $\int x\, f(x)\, dx$
- B. $\int x^2\, f(x)\, dx$
- C. $\Sigma\, x^2$
- D. $\int f(x)\, dx$

Q4. If E[X] = 2 and Var(X) = 5, find $E[X^2]$.

- A. 1
- B. 5
- C. 7
- D. 9

Q5. For the uniform PDF f(x)=1 on $0 \le x \le 1$, find E[X].

- A. 0
- B. 1/2
- C. 1/3
- D. 1

Section A – MCQs

Q1. A random variable X has the following distribution:

X	0	1	2
P(X)	0.2	0.5	0.3

Find $E[X]$.

A. 0.8

B. 1.1

C. 1.3

D. 1.5

Q2. If $E[X] = 4$ and $Var(X) = 9$, find $E[X^2]$.

A. 13

B. 16

C. 25

D. 36

Q3. If $Var(X) = 5$, find $Var(3X - 7)$.

A. 15

B. 38

C. 45

D. 52

Q4. A fair coin is tossed twice. Let X denote the number of heads. Find $Var(X)$.

A. 0.25

B. 0.5

C. 1

D. 2

Q5. A continuous random variable has PDF:

$$f(x) = 2x, \quad 0 \leq x \leq 1$$

Find $E[X]$.

A. $1/3$

B. $1/2$

C. $2/3$

D. 1

Section B – MSQs

Select all correct options.

Q6. Which statements about expectation are always true?

A. $E[X + Y] = E[X] + E[Y]$

B. $E[aX] = aE[X]$

C. $E[XY] = E[X]E[Y]$ for all random variables

D. $E[c] = c$

Q7. Which statements about variance are correct?

A. $Var(X + c) = Var(X)$

B. $Var(aX) = aVar(X)$

C. $Var(aX + b) = a^2Var(X)$

D. $Var(X) > 0$

Q8. Let X be a random variable with finite second moment. Which statements are correct?

A. $Var(X) = E[X^2] - (E[X])^2$

B. $E[X^2] \geq (E[X])^2$

C. $Var(X) = 0$ implies X is constant with probability 1

D. $E[X^2] = (E[X])^2$ always

Section C – NAT

Enter numerical answers.

Q9. A random variable X has:

$$E[X] = 5, \quad Var(X) = 4$$

Find:

$$E[(X-2)^2]$$

Q10. A continuous random variable has PDF:

$$f(x) = \begin{cases} 3x^2, & 0 \leq x \leq 1 \\ 0, & \text{otherwise} \end{cases}$$

Calculate its variance.